Predictive modelling with machine learning

Semester: Spring 2026
Course code: GRA 4160
BI Norwegian Business School

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In this course we focus on basic machine learning methods, both for supervised and unsupervised learning. The course covers both the theory and the practice of machine learning using Python.

Compulsory Readings

1. The elements of statistical learning: data mining, inference, and prediction by *Trevor J. Hastie; Robert J. Tibshirani; Jerome Friedman (Link)*

Mini project and presentation

You will work on a mini project in groups of 3 students. The mini project will be a small-scale project where you will apply the methods you learn in the course to a real-world problem.
To complete the project, you will need to perform the following steps:

- Identify a problem or research question that you would like to address.
- Collect and preprocess data for your model.
- Choose a machine learning algorithm (or several) and implement it in your project.
- All the groups can book two meetings with me to discuss the project and get some feedback either in person or on Zoom during the semester.
- Make a presentation summarizing your findings and discussing the results.
- Present for the class.

The project is an important part of the course, as it allows you to demonstrate your understanding of machine learning concepts and techniques, and to apply them in a practical setting. It is also an opportunity for you to explore a topic of interest and to develop your skills as a data scientist. The project will be graded and will count for 30% of your final grade.

Lecture Plan – Lectures are Wednesdays from 12:00–13:45.

Lecture 1: [January 7th] **Introduction and data preprocessing**

- Ch. 1 in ESL
- Notebooks: `01_Working_with_data_in_jupyter_notebooks.ipynb`
- Exercise: `01_Data_preprocessing_titanic.ipynb`

Lecture 2: [January 14th] **Machine learning basics and supervised learning**

- Ch. 2 and 6.6 in ESL
- Notebooks: `02_OLS.ipynb`, `03_Supervised_learning_with_kNN.ipynb`
- Exercise: `02_Spam_filtering_with_naive_bayes.ipynb`

Lecture 3: [January 21st] **LDA and regularized regression analysis**

- Ch. 3 and 5 in ESL
- Notebooks: `04_Linear_discriminant_analysis.ipynb`, `05_Regularised_regressions.ipynb`
- Exercise: `03_Predicting_house_prices.ipynb`

Lecture 4: [January 28th] **Classification analysis**

- Ch. 4 and 9 in ESL
- Notebooks `06_Logistic_regression.ipynb`, `07_Decision_trees.ipynb`
- Exercise: `04_Recognizing_handwritten_digits.ipynb`

Lecture 5: [February 4th] **Model selection, evaluation and assessment**

- Ch. 7 in ESL
- Notebooks: `08_IC_and_CV.ipynb`, `09_Bias_variance_tradeoff.ipynb`
- Exercise: `05_Model_selection_evaluation_and_assessment.ipynb`

Lecture 6: [February 11th] **Unsupervised learning**

- Ch. 13 and 14 in ESL
- Notebooks: `10_PCA.ipynb`, `11_K-means.ipynb`
- Exercise: `06_Text_classification.ipynb`

Lecture 7: [February 18th] **Ensemble methods and Random forests**

- Ch. 8, 10, 15 and 16 in ESL
- Notebooks `12_Introducing_ensemble_methods.ipynb`, `13_Random_forests.ipynb`
- Exercise: `07_Predicting_income.ipynb`, `08_Predicting_credit_risk.ipynb`

Lecture 8: [February 25th] **Backpropagation and gradient descent**

- Ch. 11 in ESL
- Notebooks: `16_Gradient_Decent.ipynb`, `17_Backpropagation.ipynb`
- Exercise: TBA

Lecture 9: [March 4th] **PyTorch and neural networks**

- Chapter 11 in ESL
- Notebooks: `18_NN_Basics.ipynb`
- Exercise: `09_Neural_Networks_with_PyTorch.ipynb`

Lecture 10: [March 11th] **Traditional ML vs Deep Learning**

- Ch. 11 in ESL
- Notebooks: TBA
- Exercise: TBA

Lecture 11: [March 18th] **Presentation and defense of the mini projects**

Lecture 12: [March 25th] **Summing up and Q&A**

Final exam

- Exam format: 30 hour take-home exam
- Start: 26.05.2026 at 09:00
- Submission deadline: 27.05.2026 at 15:00
- Weight: 70%

Software

In this course we will use Python. I recommend installing Python through the Anaconda distribution. You can run into some problems if you don't have the correct version of the software installed. Try to install the latest version you find. Python 2 will not be supported.

Most of the material will be in the form of Jupyter notebooks. I also recommend installing an integrated development environment (IDE) for Python. Two good options are PyCharm and VS code. Both IDEs are free for students, and they support Jupyter notebooks.

We will also rely on some third party Python libraries. I recommend spending some time familiarizing yourself with the following libraries:

1. Scikit-learn: https://scikit-learn.org/stable/
2. Pandas: https://pandas.pydata.org/
3. Matplotlib: https://matplotlib.org/
4. PyTorch: https://pytorch.org/